

Word embedding

What is Word Embedding?

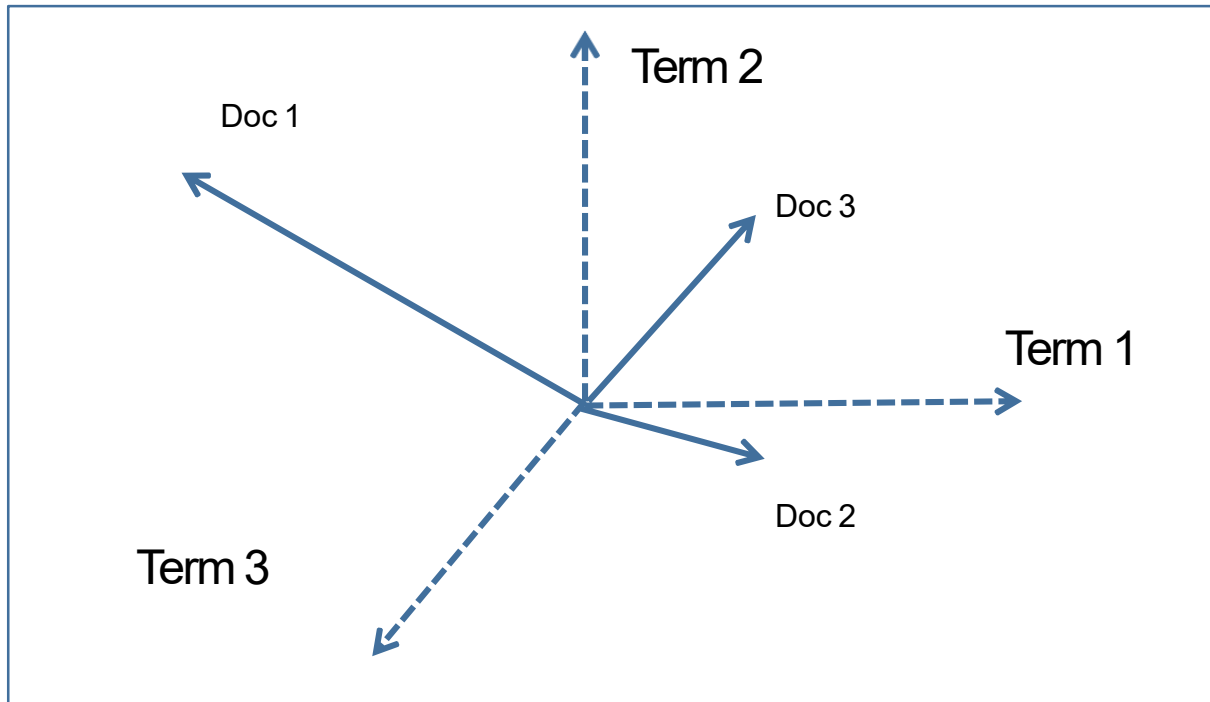
- Natural language processing (NLP) models do not work with plain text. So, a numerical representation was required.
- Word embedding is a class of techniques where word is represented as a real value vectors.
- It is a representation of word in a continuous vector space.
- It is a dense representation in a vector space.
- It can be represented in smaller dimension compared to sparse representation like one-hot encoding.
- Most of the word embedding method is based on “distributional hypothesis” by Zellig Harris.

What is word embedding? continued

- The Distributional Hypothesis is that words that occur in the same contexts tend to have similar meanings. (Harris, 1954)
- Word embeddings are designed to capture the similarity between representation like: meaning, morphology, context etc.
- The captured relationship helps us to work on downstream NLP task like chat-bot, text summarization, information retrieval etc.
- It is generated by co-occurrence matrix, dimensionality reduction and neural networks.
- It can be broadly categorized in two parts: frequency-based embeddings and prediction-based embeddings.
- The earliest work to give a vector representation was vector space model used in information retrieval task.

Vector space model

- A document was represented in a vector space.
- The dimensionality of vector space is of size of unique words in corpora.



	Term 1	Term 2	Term 3
Doc 1	0	5	5
Doc 2	2	0	1
Doc 3	3	3	0

- Hypothetical corpora with three words represented as dimension.
- Three doc projected in the vector space as per their term frequency

Vector space model continued

- The document got a numerical vector representation in a vector space represented by words.
- E.g.
 - Doc 1 -> [0, 5, 5]
 - Doc 2 -> [2, 0, 1]
- This representation is sparse in nature. Because, in real life scenario the dimensionality of a corpus shoots up to millions.
- It is based on term frequency.
- TF-IDF normalization is applied to reduce the weightage of frequent words like „the“, „are“ , and etc.
- One-hot encoding is a similar technique to represent a sentence/document in vector space.
- This representation gather limited information and fails to capture the context of the word.

Co-occurrence matrix

- It is applied to capture the neighbouring word that appeared with the word under consideration. A context window is considered to calculate co-occurrence.
- E.g.:
 - India won the match. I like the match.
 - Co-occurrence matrix for above two sentence for context window of 1.

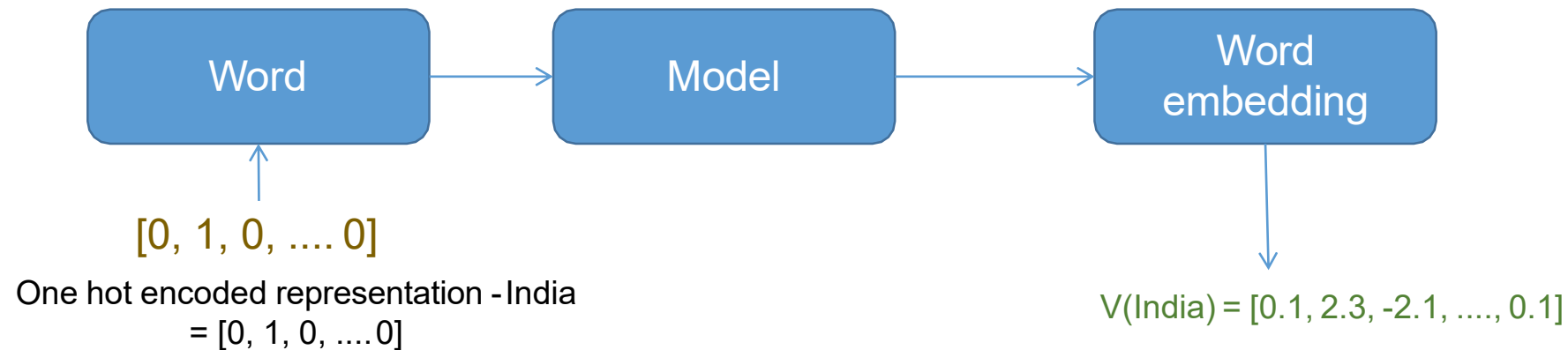
	India	won	the	match	I	like
India	1	1	0	0	0	0
won	1	1	1	0	0	0
the	0	1	1	1	0	1
match	0	0	1	1	0	0
I	0	0	0	0	1	1
like	0	0	1	0	1	1

Co-occurrence matrix continued

- Representations like One-hot encoding, Count based method and co-occurrence matrix based methods are very sparse in nature.
- Either context was limited or absent all together.
- Single representation for word in every context.
- Relation between two words like: semantic reasoning is not possible with this representation.
- Context is limited but predetermined.
- Long term dependencies are not captured.

Prediction based word embeddings

- It is a method to learn dense representation of word from a very high dimensional representation.



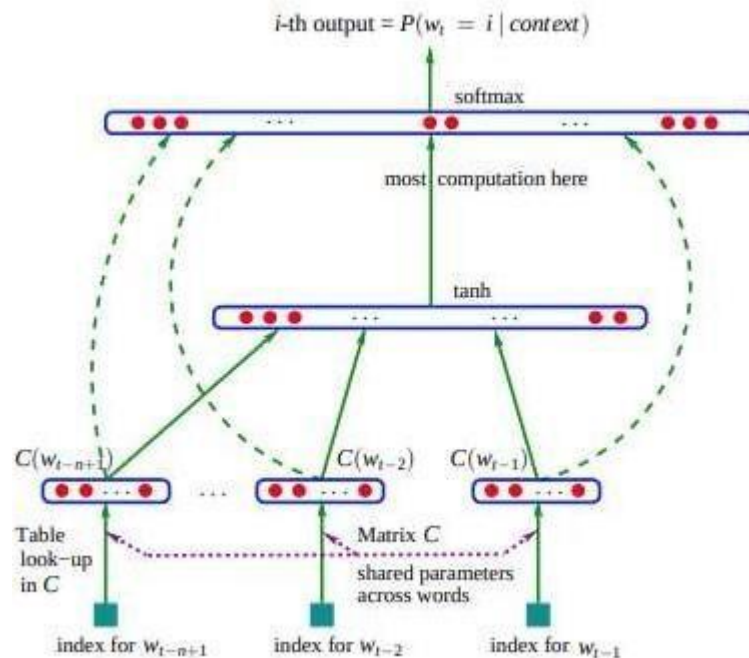
- It is a modular representation, where a sparse vector is fed to generate a dense representation

Language modelling

- Word Embedding models are very closely related to Language modelling.
- Language modelling tries to learn a probability distribution over the words in Vocabulary (V)
- Prime task of language model to calculate the probability a word W_i given the previous (N-1) words, mathematically $P(W_i|W_{i-1}, \dots, W_{i-n+1})$
- Probabilities over n-gram is calculated by frequency by constituent n-gram.
- In Neural network as well we achieve the same using softmax layer.
 - We calculate the log probability of W_i and normalize it with the sum of the probabilities over all the words.
 - $$P(W_i|W_{i-1}, \dots, W_{i-n+1}) = \frac{\exp(h^T V_{W_i}')}{\sum_{W_i \in V} \exp(h^T V_{W_i}')}$$
 - In this case, h is the representation from hidden layer and V_{W_i}' is the embedding of the word.
 - The inner product of $h^T V_{W_i}'$ generate the log probability of word W_i

Classical Neural language model

- It was proposed by Bengio et al., 2003



- It consists of one layer feed-forward neural network to predict next word in sequence.
- The model tries to maximize the probability as computed by softmax.
- Bengio $L = \frac{1}{T} \sum_t \log f(w_t, w_{t-1}, \dots, w_{t-n+1}; \theta)$ concepts
 - Embedding layer: a layer that generates word embeddings by multiplying an index vector with a word embedding matrix.

Classical Neural language model continued

- Intermediate layers: One or more layers that produce an intermediate representation of the input, e.g. a fully-connected layer that applies a non-linearity to the concatenation of word embeddings of n previous word
- Softmax Layer: the final layer that produces a probability distribution over words in V
- Intermediate layer can be replaced with LSTM.
- The network has computational complexity bottleneck due to softmax layer, in which probability over the set of vocabulary needs to be computed.
- Neural based word embedding model made a significant progress with Word2vec model proposed by Mikolov et.al. in 2013

Word2Vec

- It was proposed by Mikolov et.al. in 2013.
- It is a two layer shallow neural network trained to learn the contextual relationship.
- It places contextually similar word near to each other.
- It is a co-occurrence based model.
- Two variants of the model was proposed
 - Continuous bag of words model (CBOW)
 - Given the context word, predict the center word
 - Order of context words are not considered, so this representation is similar to BOW.
 - Skip-gram model

What does context mean?

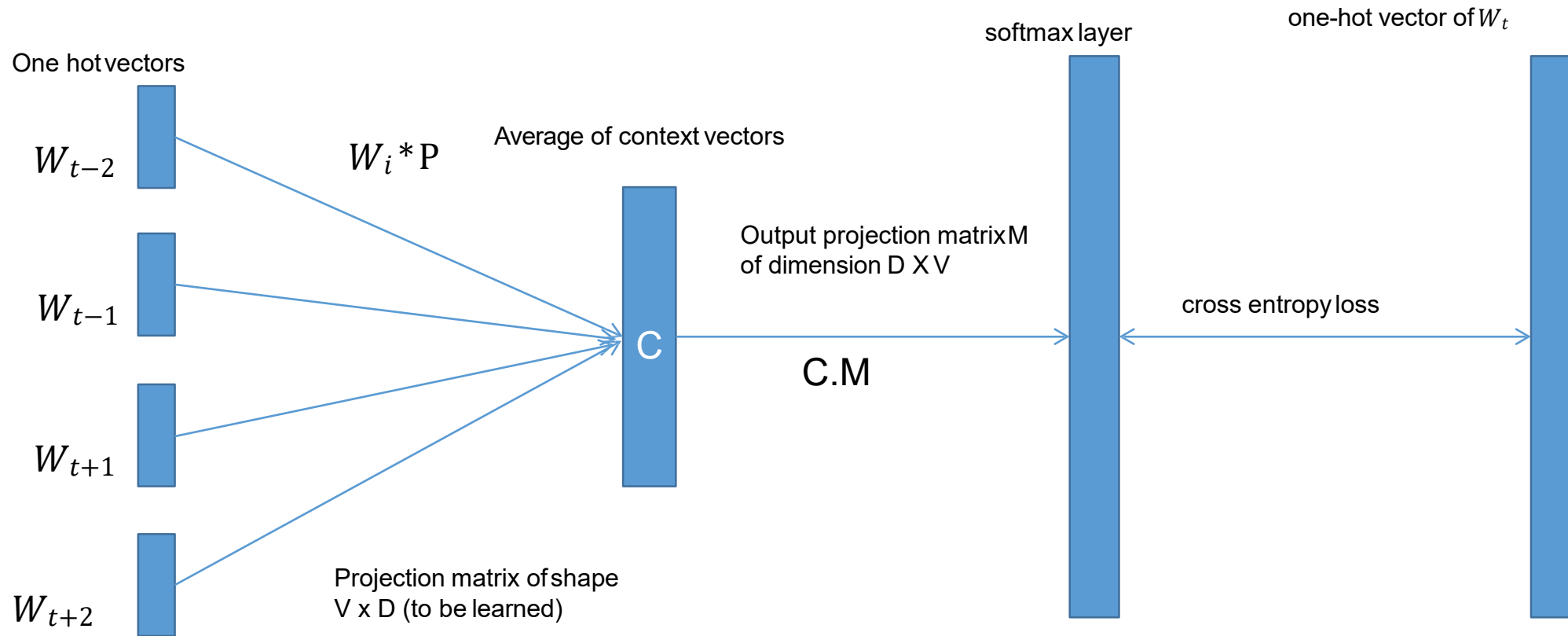
- Context is co-occurrence of the words. It is a sliding window around the word under the consideration.

India	is	now	inching	towards	a	self	reliant	state
India	is	now	inching	towards	a	self	reliant	state
India	is	now	inching	towards	a	self	reliant	state
India	is	now	inching	towards	a	self	reliant	state
India	is	now	inching	towards	a	self	reliant	state
India	is	now	inching	towards	a	self	reliant	state
India	is	now	inching	towards	a	self	reliant	state
India	is	now	inching	towards	a	self	reliant	state

Window size = 2, Yellow patches are words are in consideration, orange box are the context window

CBOW continued

- Goal: Predict the center word, given the context words.

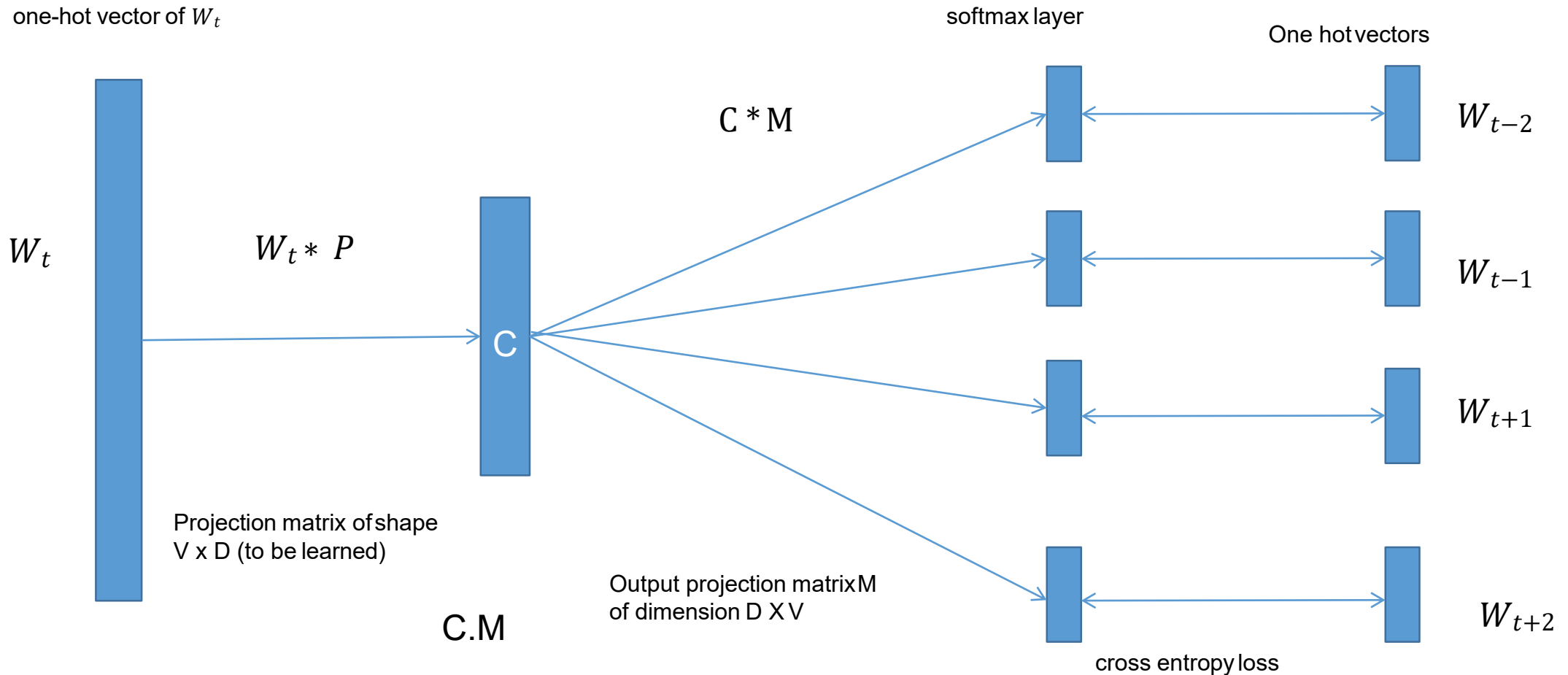


CBOW continued

- One hot encoded of the context words $W_{t-2}, \dots, W_{t+2}$ is input to the model.
- Projection matrix of shape $V \times D$, where V is the total no of unique words in the corpus and D is dimension of the dense representation, to project one-hot encoded vector into D -dimension vector.
- Averaged context vector is projected back to V -dimensional space. Softmax layer converts the representation into probabilities for W_t .
- The model is trained using cross-entropy loss between the softmax layer output and the one-hot encoded representation of W_t .

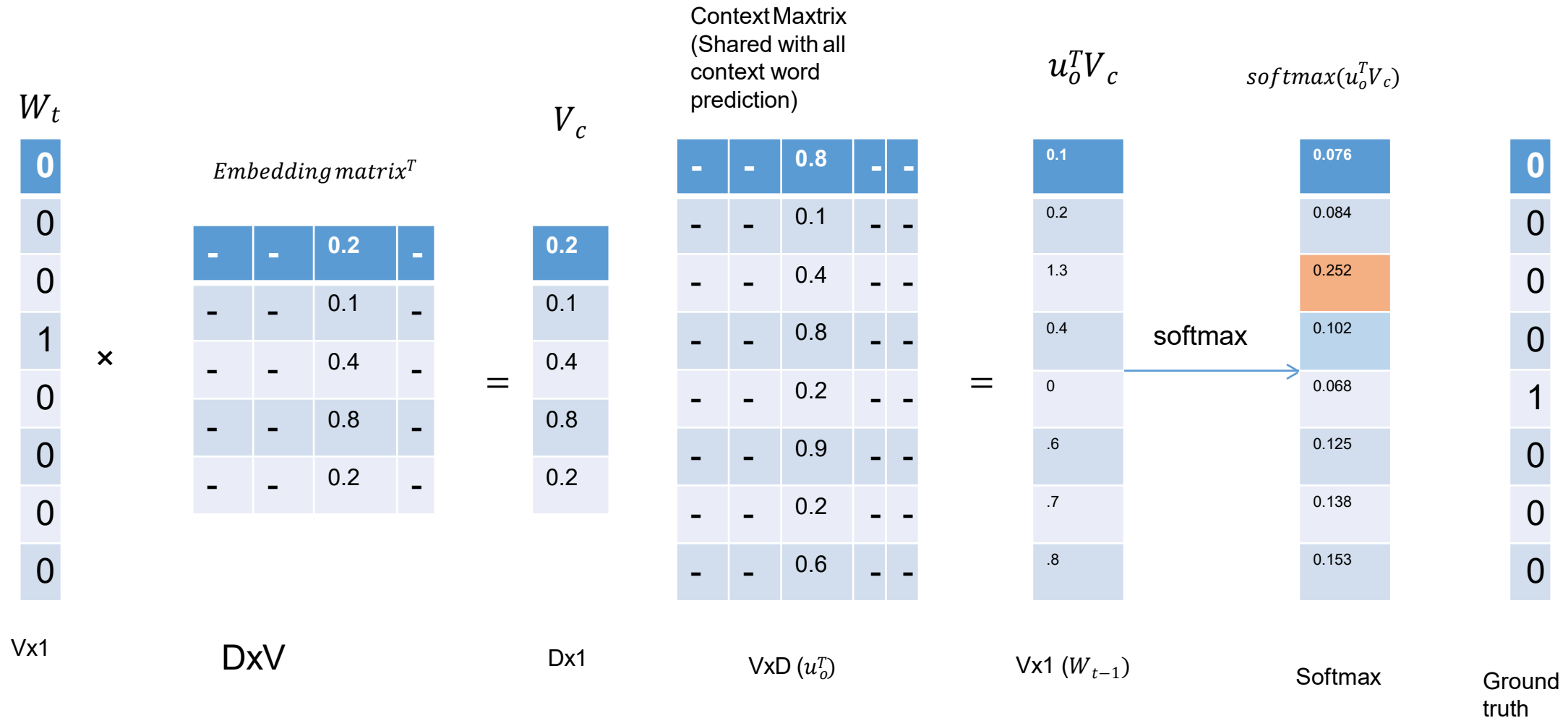
Skip-gram model: High level

- Goal: To predict the context word $W_{t-2}, \dots, W_{t+2}$ given the word W_t



Skip-gram continued

- Aend to end flow of training:



Skip-gram continued

- It focus on optimization of loss for each word:

- $P(o|c) = \frac{\exp(u_o^T v_c)}{\sum_{k=1}^V \exp(u_k^T v_c)}$

- It calculates probability of output context word given the center word c.
 - The loss function which it tries to minimize is:

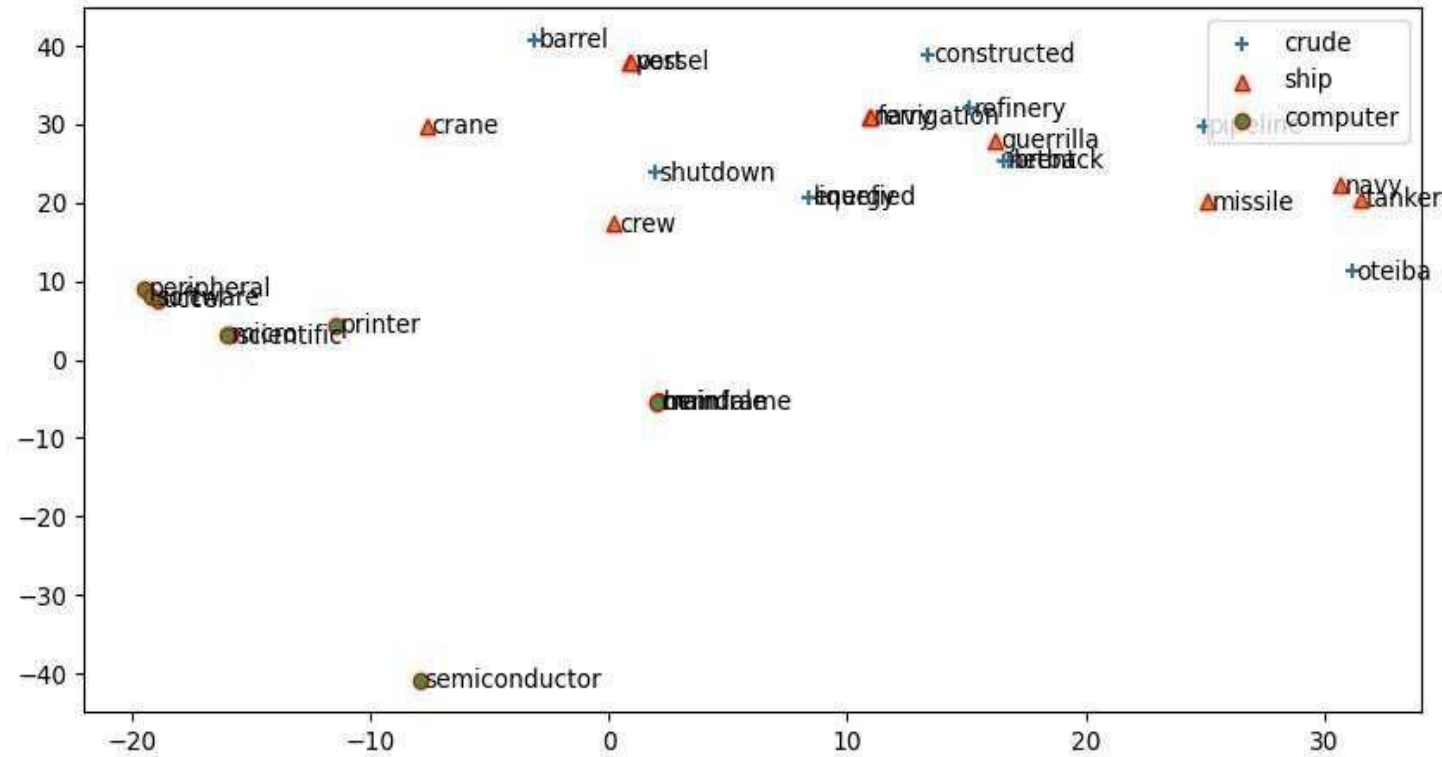
$$J(\theta) = - \frac{1}{T} \sum_{t=1}^T \sum_{\substack{-m \leq j \leq m \\ j \neq 0}} \log p(W_{t+j}/W_t)$$

- Log value is calculated similar to depicted in first equation.
 - Naive Training is costly because gradient calculation is of order $|V|$
- Two computationally efficient methods are proposed:
 - Hierarchical Softmax
 - Negative sampling

Skip-gram continued

- Use of negative sampling in training is more prevalent.
- In early example, tuple like (India, the), (India, now) are the example to true cases.
- Any corrupted tuple is called as negative sample. like (India, reliant), (India, state)
- This process with modified objective function results in a logistic regression to classify a tuple as a true combination or a corrupted one.
- The corrupt tuple is generated by sampling such that less frequent words are picked up more often as a corrupt tuple.

Word Embedding visualization



Top 5 similar words

crude barrel 0.548
crude oteiba 0.464
crude netback 0.45
crude refinery 0.438
crude pipeline 0.421

ship vessel 0.623
ship port 0.575
ship tanker 0.496
ship navigation 0.471
ship crane 0.463

computer software 0.602
computer micro 0.559
computer printer 0.542
computer mainframe 0.538
computer hemdale 0.527

- Even with a smaller corpus it can capture semantically relevant words.

t-SNE 2. D projection of Word2vec (gensim implementation) embeddings of top 10 similar words, trained for 50 epoch on Reuters news corpus from NLTK, with context len 15, vector dimension 100

Word2vec results

- The top 5 similar words when:

Context length is 30

crude barrel 0.475
crude refinery 0.438
crude stockdraws 0.427
crude yates 0.408
crude utilized 0.382

šīp vessel 0.557
ship tanker 0.506
ship port 0.5
ship icebreaker 0.461
ship loaded 0.453

čōmputer software 0.569
computer micro 0.517
computer memory 0.498
computer disk 0.495
computer printer 0.476

Embedding dimension is 300

crude refinery 0.27
crude stockdraws 0.254
crude barrel 0.244
crude utilized 0.242
crude liquefied 0.239

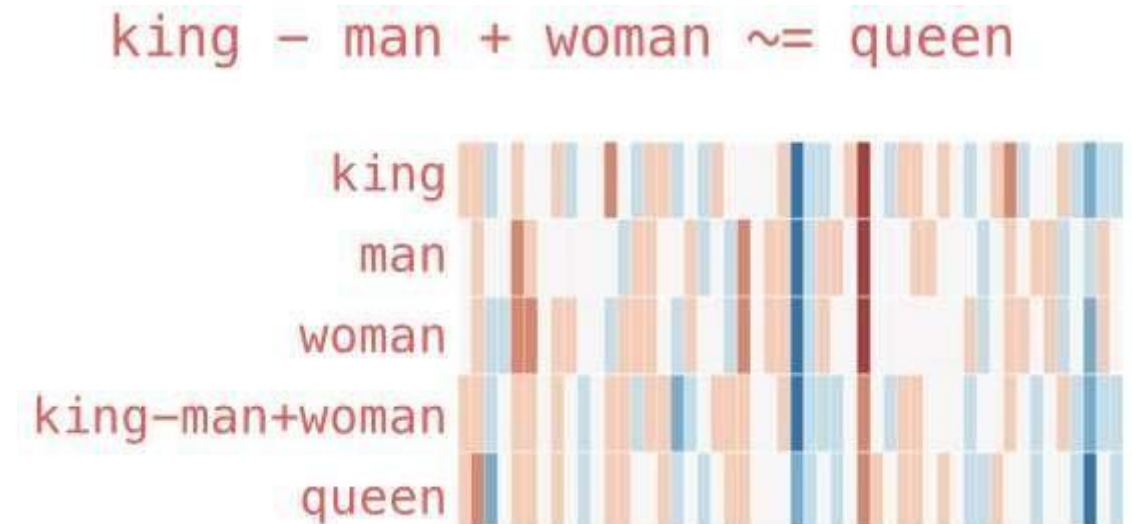
šīp vessel 0.468
ship crew 0.318
ship tanker 0.308
ship shipbuilder 0.302
ship yard 0.288

čōmputer software 0.441
computer disk 0.345
computer printer 0.345
computer uccel 0.338
computer scientific 0.335

Analogies

- Representation of analogy in vector space using word2vec vectors:
 - Vector representation of “King-man+woman” is roughly equivalent to the vector representation of queen
 - Using gensim and pretrained word2vec the analogy vector generated for “King-man+woman” 5 most closer relationships are

queen	0.7118
monarch	0.619
princess	0.5902
crown_prince	0.55
prince	0.54

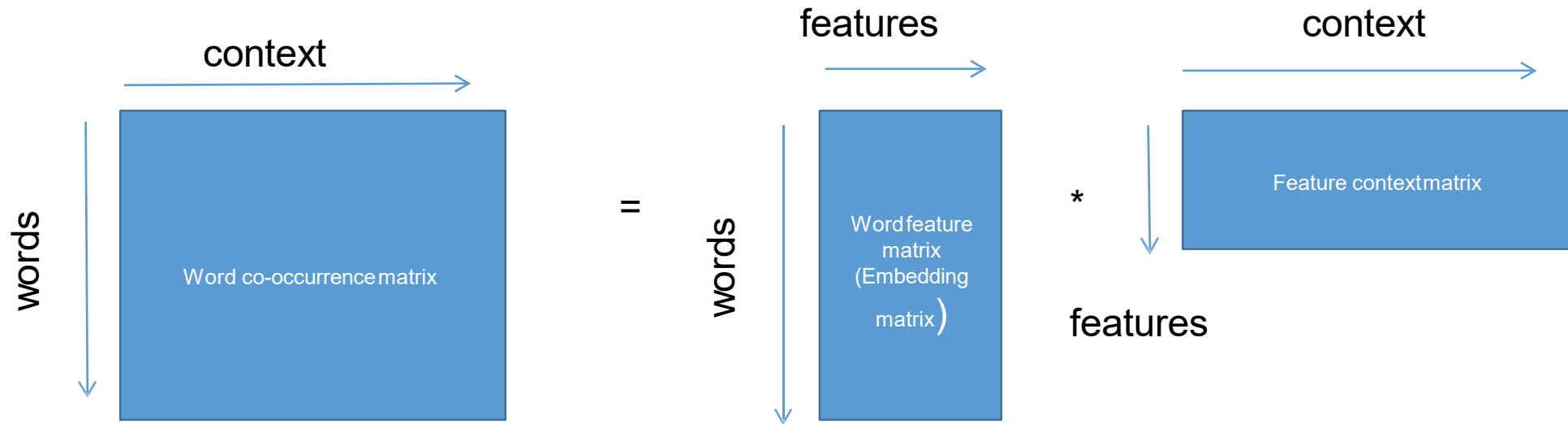


GloVe: Global Vectors for Word Representation

- It is a unsupervised learning to learn the word representation.
- It is based on co-occurrence matrix.
 - The co-occurrence matrix built on the whole corpus.
 - It is able to capture global context.
- It encompass best of the two model families:
 - Local context window method
 - Global matrix factorization
- Earlier matrix factorization like LSA was used to reduce the dimensionality.
- Two things that GloVe model captures:
 - Statistical measure using co-occurrence matrix
 - Context, by considering the neighbouring words

Glove continued

- It moves away from old matrix factorization. By considering relationship reasoning (Semantic and syntactic), GloVe tries to learn the representation for words.
- It can be represented as:



GloVe continued

- How does GloVe learns embedding?
 - It considers word-word co-occurrence probabilities as the potential of relation between words.

Probability and Ratio	$k = solid$	$k = gas$	$k = water$	$k = fashion$
$P(k ice)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k steam)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k ice)/P(k steam)$	8.9	8.5×10^{-2}	1.36	0.96

- The authors presented a relation with steam and ice as target words.
 - It is common to consider steam occur with gas and ice with solid.
 - Other co-occur words are “water” and “fashion”. “Water” has some shared property while “fashion” is irrelevant.
 - Only in the ratio of probabilities cancels out the noisy words like “water” and “fashion”.
 - As presented in the above table, the ratio of probabilities are maximum for $P(k|ice) / P(k|steam)$ is high for solid and small for gas.

GloVe continued

- What is the optimization function for GloVe?
 - In a co-occurrence matrix X the X_{ij} represents the co-occurrence count.
 - X_i is the total number of times the word appears in the context.
 - $P_{ij} = P(j/i) = X_{ij}/X_i$ is the probability of word j appear in the context of word X_i
 - For a combination of three words i, j, k . A general representation of the model is

$$F(W_i, W_j, \tilde{W}_k) = \frac{P_{ik}}{P_{jk}}$$

- The optimization function proposed by authors are:

$$J = \sum_{i,j=1}^V f(X_{ij})(W_i^T \tilde{W}_j + b_i + b_j - \log X_{ij})^2$$

Glove Continued

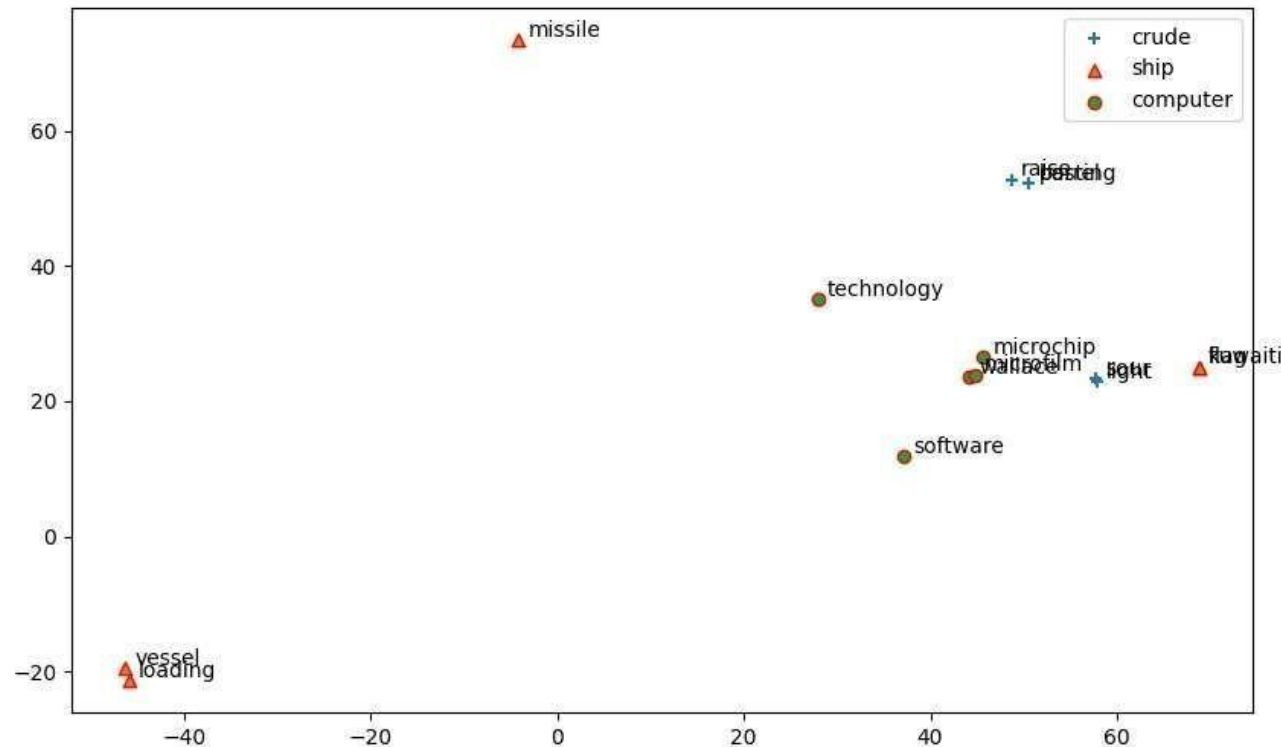
- Where V is the size of vocabulary and W_i^T and b_i is the vector and bias of the word W_i and $\tilde{W}_j$ and b_j is the context vector and its bias. The last term is the probability of occurring i in the context of j .
- The function $f(X)$ should have following properties:
 - It tends to zero at when $X \rightarrow 0$
 - It should be non-decreasing so that it can discriminate rare co-occurrence instances.
 - It should not overweight frequent co-occurrence.
- The choice of $f(X)$ is

$$f(X) = (X/X_{max})^\alpha \quad \text{if } X < X_{max} \text{ otherwise } 1$$

Glove Continued

- The model has following computational bottlenecks:
 - Creating a big co-occurrence matrix of size $V \times V$.
 - The model computational complexity depends on the number of non-zero elements.
- During training the context window needs to be sufficiently large so that the model can distinguish left context and right context.
- Words which are more distant to each other contribute less in the count. Because, distant words contribute less to the relationship of the words.
- The model generates two set of W and $\tilde{W}$. An average of both is used as the representation of words.

Glove results



t-SNE 2D projection of GloVe embeddings of top 5 similar words, trained for 50 epoch on Reuters news corpus from NLTK, with context len 15, vector dimension 100

Top 5 most relevant word list:

crude barrel0.752
crude posting0.58
crude raise 0.537
crude light0.505
crude sour0.502

ship loading0.58
ship kuwaiti 0.54
ship missile 0.537
ship vessel 0.522
ship flag 0.522

computer wallace0.595
computer software0.592
computer microfilm0.559
computer microchip0.536
computer technology0.52

Are Word2vec and GloVe enough?

- Both the embeddings can not deal with out of vocabulary words.
- Both can capture the context, but in a limited sense.
 - They always produce single embedding for the word in consideration.
 - They can't distinguish:
 - “I went to a **bank.**” and “I was standing at a river **bank.**”
 - It will always produce single representation for both the context.
- Both gives decent performance than encoding like tf-idf, count vector etc.
- Does pretrained model helps the case?
 - Pretrained models on huge corpus shows better performance compared to small corpus.
 - Pretrained models of Word2vec² is available from Google and GloVe¹ is available on Stanford's website.

1. <https://nlp.stanford.edu/projects/glove/>

2. <https://drive.google.com/file/d/0B7XkCwpI5KDYNINUTTISS21pQmM/edit?usp=sharing>

Fasttext

- It was proposed by FacebookAI team.
- It was primarily meant to handle the out of vocabulary issue of GloVe and Word2vec.
- It is an extension of Word2vec.
- This model relies on n-gram character rather than word to generate the embeddings.
- This model relies on the morphological features of a word.
- The n-gram character of a word can be represented as below:
 - For word <where> and n=3 the word n-gram characters are:
 - <wh, whe, her, ere, re>
- The final representation for word “Where” is the sum of the vector representation of <wh, whe, her, ere, re>.

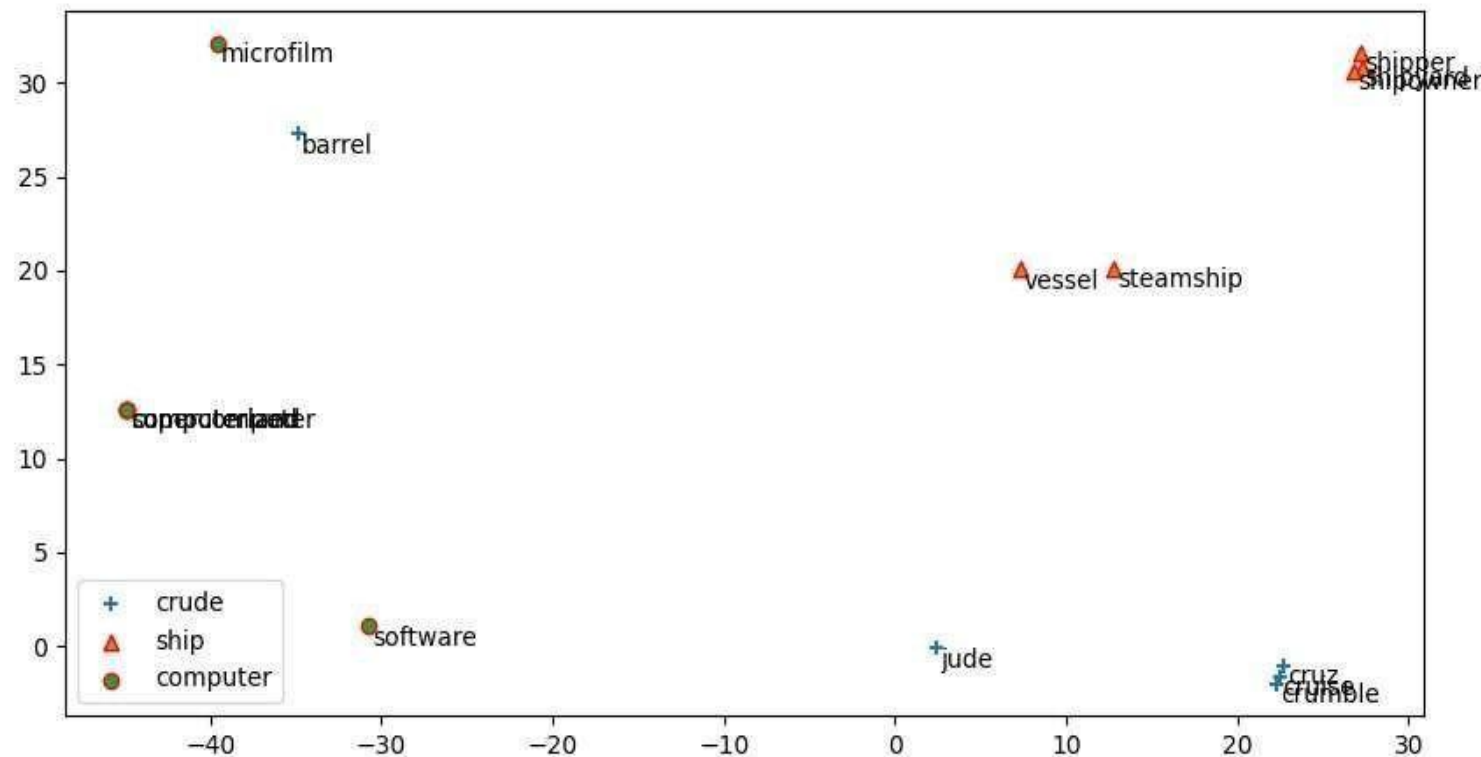
Fasttext continued

- The modified scoring function is

$$S(w, c) = \sum_{g \in G_c} Z_g^T V_c$$

- Where G_c is the set of n-grams for word w and Z_g is the vector representation of n-gram.
- The n-gram vector learning enables the model to learn the representation for out-of-vocabulary words as well.

Fasttext results



t-SNE 2D projection of fasttext embeddings (gensim implementation) of top 15 similar words, trained for 50 epoch on Reuters news corpus from NLTK, with context len 15, vector dimension 100

Top 5 most relevant word list:

crude cruz 0.582
crude barrel 0.561
crude cruise 0.501
crude crumble 0.433
crude jude 0.41

ship shipyard 0.714
ship steamship 0.703
ship shipowner 0.688
ship shipper 0.668
ship vessel 0.667

computer supercomputer 0.843
computer computerized 0.823
computer computerland 0.773
computer software 0.54
computer microfilm 0.52

Observation

- None of the representation can capture the contextual representation. Meaning that representation based on the usage.
- These are based on dictionary based look up to get the embeddings.
- Limited performance on task such as question answering, summarization compare to current state of art models like ELMO (LSTM based), BERT (transformer based) etc.

Reference:

[word2vec-Text-TensorFlow](#)